CellPress

Current Biology

Dispatches

Neurofeedback: Training the brain from within

Erica L. Busch[1,2,3]
[1]Center for Neuroscience Imaging Research, Institute for Basic Science, Suwon 16419, Republic of Korea
[2]Department of Psychology, Yale University, New Haven, CT 06510, USA
[3]Wu Tsai Institute, Yale University, New Haven, CT 06510, USA
Correspondence: erica.busch@yale.edu
https://doi.org/10.1016/j.cub.2026.09.065

The hippocampus is a key brain structure linked to how we encode and reconstruct our own experiences. Neurofeedback training can help people identify strategies to successfully drive their own hippocampal activity, possibly promoting stronger encoding beyond the training session.

Learning often involves figuring out how to link actions with desired outcomes. Sometimes this link is simple: swing the tennis racquet, watch where the ball lands — the connection is immediately observable, letting us adjust our form for the next shot. But what about actions whose outcomes are hidden from view? Consider, for example, a thought: the action of thinking lets us recall the past or imagine the future, but we cannot observe how a thought will affect activity in particular brain regions. Without that information, it is difficult to identify which thoughts most effectively engage a particular brain region, let alone learn to regulate that region volitionally. Neurofeedback offers a way to make this hidden action–outcome link visible, giving people real-time information about how their own brain responds to internally generated mental actions[1]. This possibility is especially intriguing for the hippocampus, a structure engaged almost entirely by internal states and crucial for reconstructing the past, planning ahead, and navigating the world[2–4]. A new study by van de Langenberg, Gurevitch *et al.* reported in this issue of *Current Biology* shows that people can use neurofeedback to discover which mental strategies most effectively drive activity in their own hippocampus[5].

The hippocampus plays a central role in generating and using internal representations of experience that extend beyond the present moment. Located deep in the medial temporal lobe with extensive brain-wide connections, it supports the encoding and organization of continuous experience into distinct, memorable episodes[3]. It is also a valuable yet vulnerable structure: the hippocampus is highly susceptible to stress and aging[6], and its dysfunction contributes to intrusive thoughts and disintegrated memories in many psychiatric populations[7]. Since the hippocampus is primarily engaged by internally generated states, a tool for strengthening or protecting its function that works through the mind's own operations might offer a route toward self-regulating the hippocampus according to its preferred pathways.

Prior work has shown that neurofeedback can train people to modulate other regions like the amygdala, a region reliably engaged by prompting participants to recall emotionally charged events[8]. The hippocampus offers no such well-defined route: it isn't as reliably triggered by external prompts but is engaged more so through internally generated processes like recall and imagination, leaving little to shape a participant's search. Despite this difficulty, recent work shows hippocampal representations can shift incidentally during more distributed neurofeedback training[9,10], hinting it is a promising, if underexplored, target in its own right.

Neurofeedback can target plasticity in deep brain regions, like the hippocampus, by measuring brain activity within the region and presenting that activity back to participants in the scanner in real time, offering them a representation of the neural signal that they can then try to change. By presenting participants with a representation of their brain activity and incentives to alter it, neurofeedback turns an otherwise invisible outcome of thoughts — brain activity — into something observable[1] (Figure 1). This process closes the loop: much as a tennis player can experiment with different stances and watch where each shot lands, neurofeedback participants can test different mental strategies and observe how each one moves the signal[11,12]. Through trial and error, participants can gradually find a strategy that reliably works — not just activating the target region once, but learning a consistent mapping between a specific mental action and its neural outcome.

Across two studies, van de Langenberg, Gurevitch *et al.* tested this directly. In Study 1, participants were told explicitly that their goal was to increase hippocampal activation and were specifically told to prepare the strategy of recalling personal memories. During neurofeedback, they recalled these memories and watched a display reflecting how much their hippocampal activity rose, with greater increases yielding more rewarding feedback. Participants succeeded at the task from the first run, establishing that people can voluntarily increase their hippocampal activity given explicit strategies. But because they were told both what to do and what outcome to aim for, an important question remained: could people identify an effective action on their own, through trial and error, without being told what the target even was?

Study 2 posed a sharper test. Participants were left blind to both the desired outcome (increased hippocampal activity) and the actions that might drive it. They were given a loose list of strategies to try — some plausibly related to the hippocampus, like recalling a memory or imagining a place, others not, like arithmetic or spelling — with no indication

All rights are reserved, including those for text and data mining, AI training, and similar technologies.

of which, if any, were expected to work. Unbeknownst to them, participants were also split into two groups that appeared identical, receiving the same instructions and feedback display. Yet, the difference between the groups was that one group's feedback reflected their own real hippocampal activity (real feedback); the other group's was recycled from a different participant's earlier session and bore no relationship to their own brain activity at all (sham feedback). This design isolated a specific question: is feedback tied to a person's own brain actually necessary for learning hippocampal control, or would motivated searching alone succeed? Since both groups searched under identical conditions but only one group's search was tied to their own hippocampal activity, differences in learning could be attributed specifically to that missing action–outcome link.

Both groups gravitated toward hippocampus-relevant strategies, like recalling a memory, at similar rates — so simply choosing a sensible category of action didn't distinguish them. The difference emerged a layer deeper: only in the real-feedback group did using these strategies predict performance, and this link grew stronger over the course of training. Sham participants, despite trying the same strategies just as often, showed no relationship between strategy attempts and performance.

This suggests that real feedback did more than point people to the right category of strategy — it helped them pin down the most effective strategy to execute within that category. Returning to the tennis analogy: many players might converge on a forehand as the right category of shot, but figuring out the specific depth, spin, and follow-through that works for their own swing requires experimenting and watching where each attempt lands. Consistent with this, some real-feedback participants converged on unique strategies that were never suggested to them at all, while sham participants' strategies stayed close to the original examples regardless of how well they performed. Feedback, therefore, may work less by teaching people what to try, and more by helping them recognize, through trial and error, when they've found a strategy that actually works.

Does this learning persist beyond the training session itself? In a final analysis,

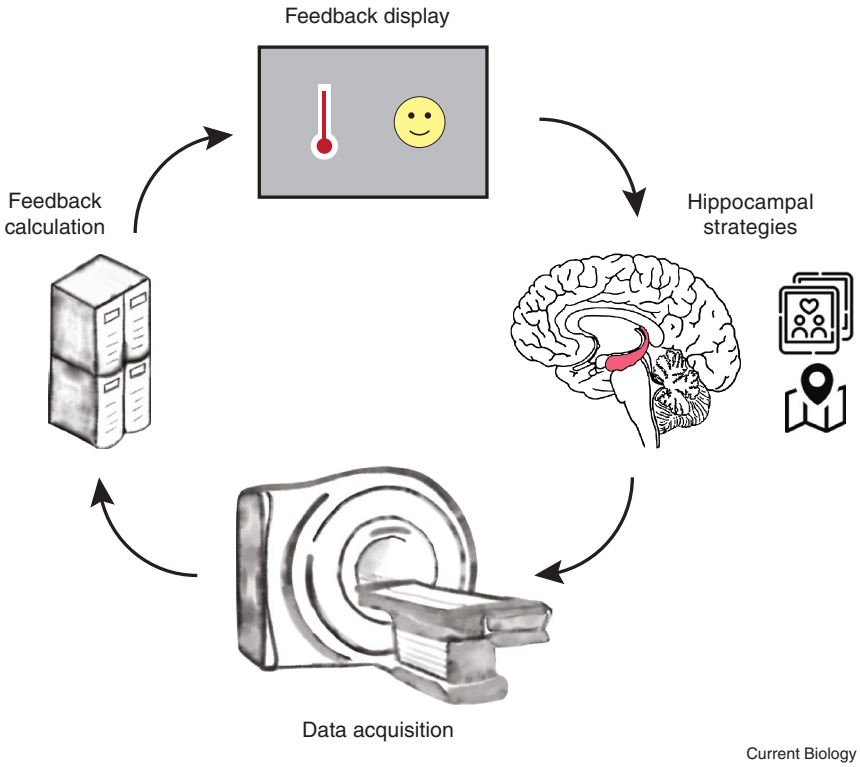


Figure 1. Neurofeedback pipeline.
In a real-time functional magnetic resonance imaging (fMRI) neurofeedback study, participants lie in an MRI scanner while images of their brain are acquired. These images are processed and a feedback score is calculated, reflecting the change in the participant's brain activity in the target region (here[5], the hippocampus). That score is presented back to participants as a change in the display, such as a thermometer becoming hotter or a face smiling, which indicates to participants how well they performed the task. This loop is then repeated over many trials of training. In some experiments (like Study 1), participants are instructed to use specific strategies, such as autobiographical recall, to change their brain activity. In other experiments (like Study 2), participants are instructed to generate their own strategies.

Study 2 participants watched a movie in the scanner after neurofeedback training. One of the hippocampus's roles is parsing continuous experience into memorable chunks, often studied by looking at how neural activity patterns shift at 'event boundaries' — moments like a scene change[13,14]. Participants who learned hippocampal upregulation over the course of neurofeedback training later showed more distinct shifts in neural activity patterns at event boundaries while passively viewing a movie (that is, without a task or feedback). Notably, more distinct neural activity at event boundaries did not correspond with hippocampal upregulation scores from the first run of neurofeedback but emerged with continued training, suggesting the relationship reflects something learned rather than a fixed trait.

Still, the design cannot disentangle whether neurofeedback training caused the event boundary effect as opposed to other confounding effects. It's also not yet known whether hippocampal upregulation training improves memory itself — a more clinically meaningful outcome the study did not directly test. And the precise mental content that made a strategy work remains unclear; participants could describe roughly what they were doing but not the fine-grained 'how', so whether real and sham participants' strategies differed in kind, not just effectiveness, remains anecdotal.

Despite these open questions, this study marks a meaningful step: it shows that people can learn to influence one of the brain's most inaccessible structures, not through precise instructions, but by being given access to the link between their thoughts and neural outcomes. That

distinction may matter beyond the hippocampus. A recurring problem across brain-based interventions is that the same prescribed approach works for some people and not others; depression, for instance, includes biologically distinct subtypes, which helps explain why a therapy effective for one patient can fail for another[15]. Current approaches in precision psychiatry mostly address this using biological profiling to sort patients into existing treatments. This study suggests an additional route: instead of matching people to a fixed set of techniques, neurofeedback could let each person discover, through their own search, the version of a strategy that works for their own brain[16] — a personalized intervention generated rather than selected.

DECLARATION OF INTERESTS

The author declares no competing interests.

REFERENCES

1. Sitaram, R., Ros, T., Stoeckel, L., Haller, S., Scharnowski, F., Lewis-Peacock, J., Weiskopf, N., Blefari, M.L., Rana, M., Oblak, E., *et al*. (2017). Closed-loop brain training: the science of neurofeedback. Nat. Rev. Neurosci. *18*, 86–100. https://doi.org/10.1038/nrn.2016.164.

2. Schacter, D.L., Addis, D.R., and Buckner, R.L. (2007). Remembering the past to imagine the future: the prospective brain. Nat. Rev. Neurosci. *8*, 657–661. https://doi.org/10.1038/nrn2213.

3. Eichenbaum, H. (2017). On the integration of space, time, and memory. Neuron *95*, 1007–1018. https://doi.org/10.1016/j.neuron.2017.06.036.

4. Bellmund, J.L.S., Gärdenfors, P., Moser, E.I., and Doeller, C.F. (2018). Navigating cognition: spatial codes for human thinking. Science *362*, eaat6766. https://doi.org/10.1126/science.aat6766.

5. van de Langenberg, S.C.N., Gurevitch, G., Markovits, T., Singer, N., Ben-Yakov, A., Hendler, T., and Schwabe, L. (2026). Real-time fMRI feedback guides self-generated cognition toward upregulation of hippocampal activity. Curr. Biol. *36*, 5429–5445.e6.

6. Lupien, S.J., McEwen, B.S., Gunnar, M.R., and Heim, C. (2009). Effects of stress throughout the lifespan on the brain, behaviour and cognition. Nat. Rev. Neurosci. *10*, 434–445. https://doi.org/10.1038/nrn2639.

7. Brewin, C.R., Gregory, J.D., Lipton, M., and Burgess, N. (2010). Intrusive images in psychological disorders: characteristics, neural mechanisms, and treatment implications. Psychol. Rev. *117*, 210–232. https://doi.org/10.1037/a0018113.

8. Herwig, U., Lutz, J., Scherpiet, S., Scheerer, H., Kohlberg, J., Opialla, S., Preuss, A., Steiger, V.R., Sulzer, J., Weidt, S., *et al*. (2019). Training emotion regulation through real-time fMRI neurofeedback of amygdala activity. NeuroImage *184*, 687–696. https://doi.org/10.1016/j.neuroimage.2018.09.068.

9. Peng, K., Wammes, J.D., Nguyen, A., Iordan, C.R., Norman, K.A., and Turk-Browne, N.B. (2024). Inducing representational change in the hippocampus through real-time neurofeedback. Philos. Trans. R. Soc. Lond. B Biol. Sci. *379*, 20230091. https://doi.org/10.1098/rstb.2023.0091.

10. Busch, E.L., Fincke, E.C., Lajoie, G., Krishnaswamy, S., and Turk-Browne, N.B. (2026). Human learning of noninvasive brain–computer interfaces via manifold geometry. Nat. Neurosci. *29*, 2429–2437. https://doi.org/10.1038/s41593-026-02311-2.

11. Birbaumer, N., Ruiz, S., and Sitaram, R. (2013). Learned regulation of brain metabolism. Trends Cogn. Sci. *17*, 295–302. https://doi.org/10.1016/j.tics.2013.04.009.

12. Lubianiker, N., Paret, C., Dayan, P., and Hendler, T. (2022). Neurofeedback through the lens of reinforcement learning. Trends Neurosci. *45*, 579–593. https://doi.org/10.1016/j.tins.2022.03.008.

13. Baldassano, C., Chen, J., Zadbood, A., Pillow, J.W., Hasson, U., and Norman, K.A. (2017). Discovering event structure in continuous narrative perception and memory. Neuron *95*, 709–721.e5. https://doi.org/10.1016/j.neuron.2017.06.041.

14. Ben-Yakov, A., and Henson, R.N. (2018). The hippocampal film editor: sensitivity and specificity to event boundaries in continuous experience. J. Neurosci. *38*, 10057–10068. https://doi.org/10.1523/JNEUROSCI.0524-18.2018.

15. Drysdale, A.T., Grosenick, L., Downar, J., Dunlop, K., Mansouri, F., Meng, Y., Fetcho, R.N., Zebley, B., Oathes, D.J., Etkin, A., *et al*. (2017). Resting-state connectivity biomarkers define neurophysiological subtypes of depression. Nat. Med. *23*, 28–38. https://doi.org/10.1038/nm.4246.

16. Zhang, K., Trambaiolli, L., Zhao, Z., Taschereau-Dumouchel, V., and Feusner, J.D. (2026). Neurofeedback interventions for obsessive-compulsive and related disorders: current evidence and future directions. J. Psychiatr. Res. *98*, 1–12. https://doi.org/10.1016/j.jpsychires.2026.03.013.

Deep-sea biology: Sisyphus on the bottom of the world

Paul V.R. Snelgrove
Department of Ocean Sciences and Biology Department, Memorial University of Newfoundland, St. John's, NL A1C 5S7, Canada
Correspondence: psnelgrove@mun.ca
https://doi.org/10.1016/j.cub.2026.09.037

In Earth's most remote environments, at the bottom of the ocean, mineral-rich nodules litter the sea floor, attracting mining exploration. How do these nodules not get buried by sediment? New evidence suggests that small invertebrates living the fate of Sisyphus may provide an answer.

The demand for important minerals such as nickel, cobalt, copper, and manganese has escalated interest in deep-sea mining of polymetallic nodules from the deep abyssal plains that cover the ocean floor at depths from 4000–7000 meters, far from most land masses[1]. Aside from curiosity about the biological complexity and potential impact of mining these remote and sensitive environments[2], scientists have long been fascinated by the provenance of these roughly potato-sized, mineral-rich ores, also known as 'manganese nodules'. This fascination focuses on understanding how nodules form[3,4], but also the role they play in creating habitat for a richly diverse but numerically sparse seafloor fauna of invertebrates that manage to survive in this severely food-limited environment[5–7]. Furthermore,

All rights are reserved, including those for text and data mining, AI training, and similar technologies.